Transmission Lines and Electromagnetics - Lecture Notes

1. Transmission Line Basics

A transmission line is characterised by its characteristic impedance Z0 = sqrt(L/C), where L and C are the inductance and capacitance per unit length. Coaxial cable commonly uses Z0 = 50 ohms; twisted pair used for broadcast television uses 75 ohms.

The velocity of propagation is v = 1/sqrt(L*C) and is often expressed as a fraction of the speed of light c = 3e8 m/s. The wavelength on the line is lambda = v/f.

2. Reflections and Matching

When a line is terminated in a load ZL that is not equal to Z0, part of the incident wave reflects. The reflection coefficient is Gamma = (ZL - Z0)/(ZL + Z0). The Voltage Standing Wave Ratio is VSWR = (1 + |Gamma|)/(1 - |Gamma|); a perfectly matched line has VSWR = 1.

Maximum power transfer occurs when ZL equals the complex conjugate of the source impedance. A quarter-wave transformer of impedance Z0' = sqrt(Z0*ZL) matches a load to a line.

3. Maxwell and Propagation

Maxwell's equations describe how electric and magnetic fields are generated and altered. In free space electromagnetic waves travel at c = 3e8 m/s. The intrinsic impedance of free space is approximately 377 ohms. The power density of a plane wave falls as 1/r^2 with distance (the inverse square law).